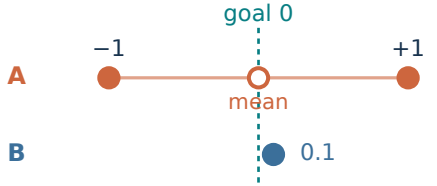
a Goal scoring

Perfect one-step mean, wrong action



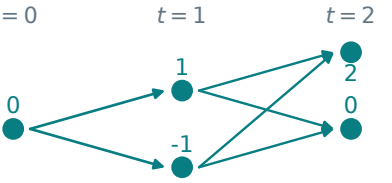
Squared goal loss

	At mean	Expected
A	0 ←	1
B	0.01	0.01 ←

Mean score selects A.
Expected loss favors B.

b Recursive rollout

$x_{t+1} = x_t^2 + \xi_t, \quad \xi_t = \pm 1$ equally



True mean: $\mathbb{E}[x_2] = 1$

Mean predictor $f(x) = x^2$

$f(f(0)) = 0 \neq \mathbb{E}[x_2]$